Advanced Data Analysis and Machine Learning (Project 1c)

— **NASA Turbofan Jet Engine** (level A): *Forecast when the engine is due to maintenance* —

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Communication

We have established a WhatsApp group for communication among the project members and decided to occasionally meet in person on campus. The code is shared using a GitHub repository — https://github.com/nedelamatej/adaml-kul.

Exploratory Analysis

The *NASA Turbofan Jet Engine* dataset consists of four subsets (`FD001` – `FD004`) each with a training set, a test set and a vector of true remaining useful life (RUL) values, distinguished by operating conditions and fault modes, as shown in Table 1. Training data were loaded into Python as four matrices (20631, 53759, 24720, 61249 observations as rows and 26 predictors as columns). All further exploratory analysis in this report focuses on `FD001` (single condition, single fault mode) for clarity. However, the same steps will also be applied to the other subsets.

Subset	Operating condition	Fault modes	Units	Rows
`FD001`	Single (sea level)	Single (HPC degradation)	100	20631
`FD002`	Six (discrete)	Single (HPC degradation)	260	53759
`FD003`	Single (sea level)	Two (HPC degradation and fan degradation)	100	24720
`FD004`	Six (discrete)	Two (HPC degradation and fan degradation)	249	61249

Table 1: *NASA Turbofan Jet Engine* dataset subsets (all of them) and their characteristics.

Each subset has 26 columns, starting with unit number, time (cycle), 3 operational settings and 21 sensor measurements, as shown in Table 5, with symbols and physical meaning taken from *Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation* (Saxena et al., 2008). The 3 operational settings (altitude, Mach number, TRA) take only 6 discrete combinations and could be treated as a single categorical attribute for operating condition.

It has been verified that there are no missing values and that each unit's time series is continuous without any gaps. However, the units have varying lifetimes, as shown in histogram in Figure 1. Columns `T2`, `P2`, `Nf_dmd` and `PCNfR_dmd` have zero standard deviation within each operating condition and therefore are redundant for modelling. Other sensor scales differ by orders of magnitude and require normalization or standardization before modelling.

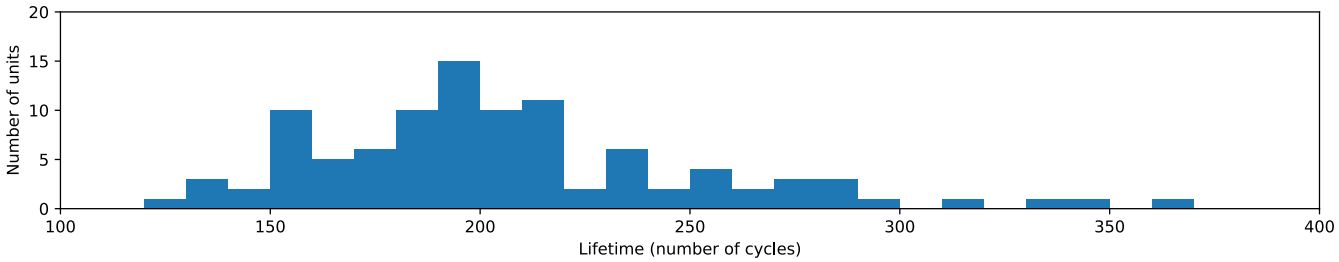

Figure 1: Histogram of unit lifetimes for `FD001`.

Given the two fault modes (HPC and fan degradation), sensors expected to be most informative for RUL include `T30`, `P30`, `Ps30` (HPC outlet conditions), `Nf` and `NRf` (fan speed). Summary statistics and plots for the `FD001` HPC outlet sensors are shown in Table 2 and Figure 2. It can be observed that the total temperature and static pressure at HPC outlet rise over the unit's lifetime.

Symbol	Description	Unit	Mean	Std. dev.	Min.	Max.
`T30`	Total temperature at HPC outlet	°R	1590.52	6.13	1571.04	1616.91
`P30`	Total pressure at HPC outlet	psia	553.37	0.89	549.85	556.06
`Ps30`	Static pressure at HPC outlet	psia	47.54	0.27	46.85	48.53

Table 2: *NASA Turbofan Jet Engine* dataset predictors (three of them) and their statistics.

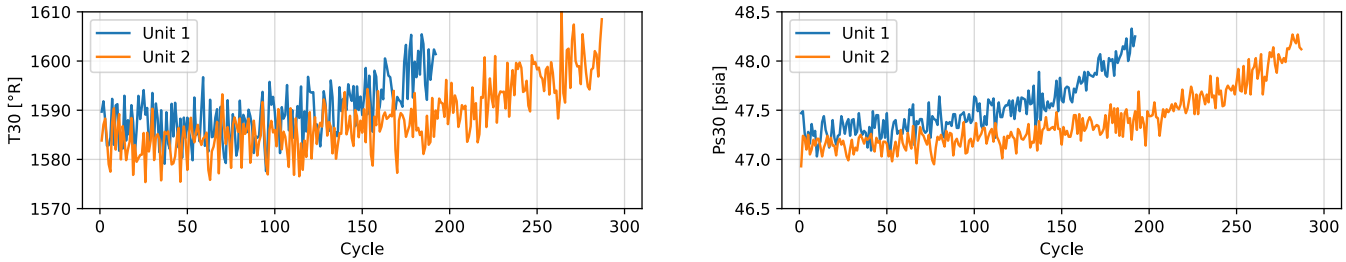

Figure 2: Total temperature (`T30`) and static pressure (`Ps30`) at HPC outlet for `FD001`.

Principal Component Analysis

The principal component analysis was applied to all four subsets. Before applying the principal component analysis on the dataset, all sensor features were standardized to eliminate the scale difference and to ensure equal contribution to the variance calculation. After applying the principal component analysis, we took 2 principal components (PC) which reduced the high-dimensional complexity of the dataset but captured a substantial share of the total variance, though considerably more for `FD002` and `FD004` (both 97 %) than for `FD001` and `FD003` (74 % and 76 %, respectively). The high variance captured for `FD002` and `FD004` likely reflects the added variation from six operating conditions rather than degradation alone, motivating the per-condition normalization planned below.

The biplots (Figure 3) and loading plots (Figure 4) clearly visualize the group of 21 sensors into distinct clusters based on their behaviors telling us which physical parameters (such as temperatures, pressures and speeds) share strong relation. Comparing the subsets highlights how operating environments change engine behavior. Datasets like `FD001` and `FD003` show smooth and unified degradation trajectories because they operate under constant conditions whereas `FD002` and `FD004` show multi-branched or scattered clusters due to varying altitude, Mach number and throttle settings. The loading vectors map out physical subsystems behavior, closely aligned sensor arrows (such as specific temperature and pressure pairings) indicating that thermal and aerodynamic subsystems tend to degrade together.

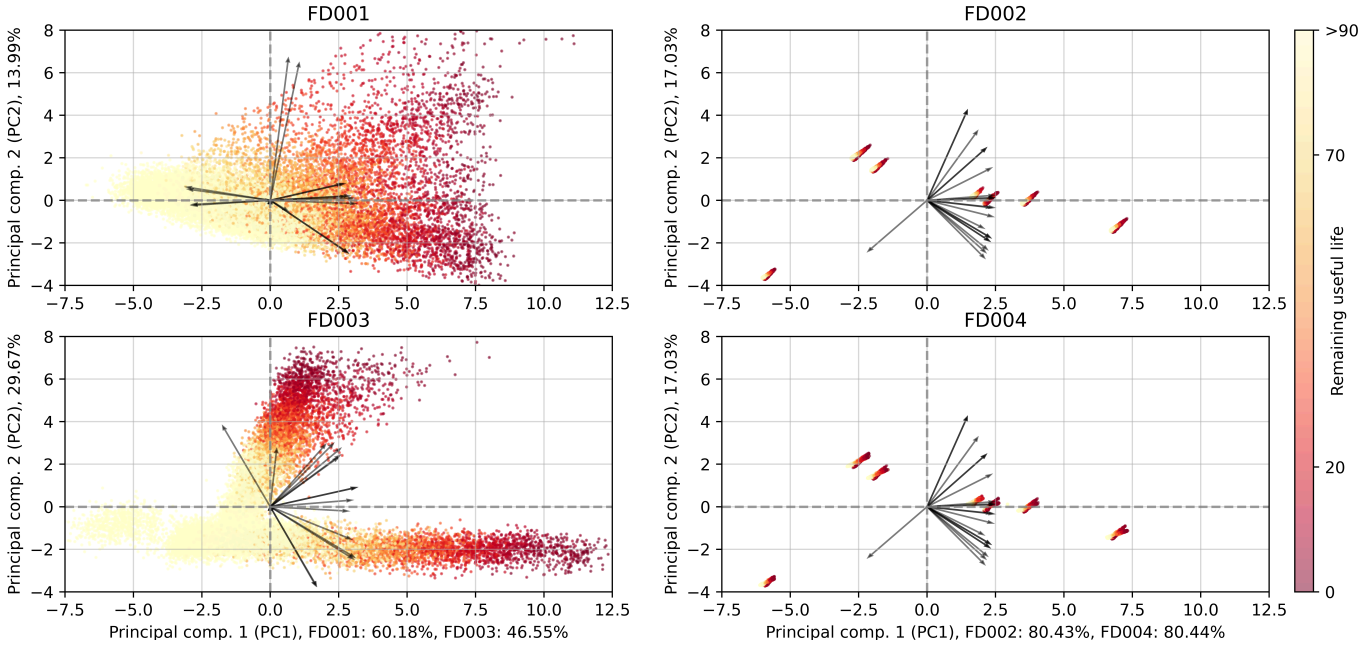

Figure 3: PCA biplots before data pretreatment for all *NASA Turbofan Jet Engine* subsets.

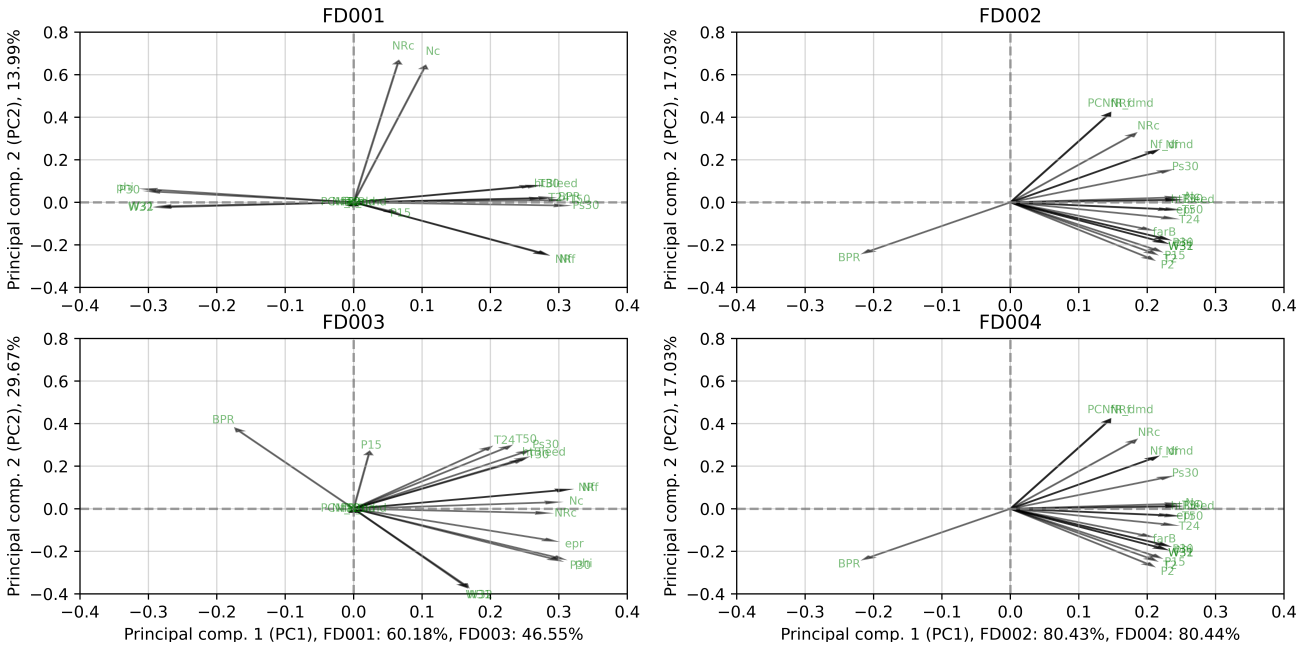

Figure 4: PCA loading plots before data pretreatment for all *NASA Turbofan Jet Engine* subsets.

Data Pretreatment

Compared to the initial plan, standardization per operational condition is applied to all subsets, not only `FD002` and `FD004`. Predictors are dropped based on their standard deviation rather than a fixed list, which additionally removes `epr` and `farB`, leaving 15 of 21 sensors. Remaining useful life capping and rolling mean smoothing have also been added.

Data pretreatment starts with rounding the 3 operational settings to remove noise and replacing these columns with one categorical variable representing the operational conditions. The lookup table of operational conditions is provided in Table 3. Operational settings are dropped from the dataset after this transformation, as the information is no longer needed and is preserved in the new categorical variable.

Operating condition	Altitude [K ft]	Mach number	TRA	Note
`Op. cond. 0`	0.0	0.00	100.0	*All subsets `FD001` – `FD004`*
`Op. cond. 1`	10.0	0.25	100.0	*Only subsets `FD002` and `FD004`*
`Op. cond. 2`	20.0	0.70	100.0	*Only subsets `FD002` and `FD004`*
`Op. cond. 3`	25.0	0.62	60.0	*Only subsets `FD002` and `FD004`*
`Op. cond. 4`	35.0	0.84	100.0	*Only subsets `FD002` and `FD004`*
`Op. cond. 5`	42.0	0.84	100.0	*Only subsets `FD002` and `FD004`*

Table 3: Lookup table of operating conditions.

After the operational conditions have been encoded as categorical variables, the remaining useful life (RUL) for each engine is calculated as shown in Equation 1, where u denotes the unit index, t the cycle number, and $t_{\max}(u)$ the unit's last recorded cycle. Remaining useful life is capped at 125 cycles. Early in its life, an engine operates in a healthy state and degradation is not yet observable in the sensor measurements (see Figure 2 and Figure 8), so the data cannot distinguish RUL 200 from RUL 150. We focused on accurate predictions at low RUL, which are more relevant from a maintenance perspective, so a capped value is interpreted as "healthy". Model predictions are therefore clipped to $[0, 125]$ and a prediction of 125 means healthy state with RUL ≥ 125. The true RUL of the test units is capped in the same way for evaluation.

$$\mathrm{RUL}(u, t) = \min(t_{\max}(u) - t, 125) \tag{1}$$

As can be seen in Figure 5, the distribution of key sensors before standardization shows significant variability and magnitude differences across sensors. Closer investigation of individual sensors (e.g. `Ps30` in Figure 6) reveals that measurements vary significantly across different operating conditions and are therefore clustered into distinct groups corresponding to the operating condition.

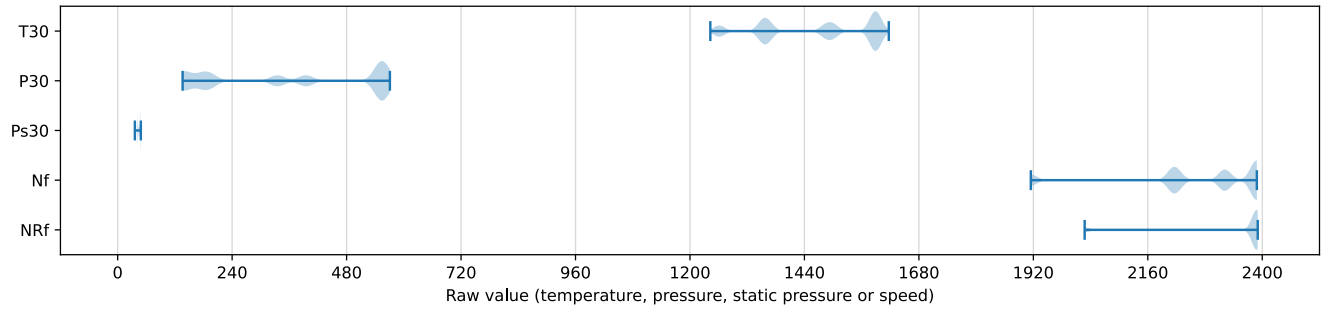

Figure 5: Violin plot of key sensors (`T30`, `P30`, `Ps30`, `Nf` and `NRf`) before standardization.

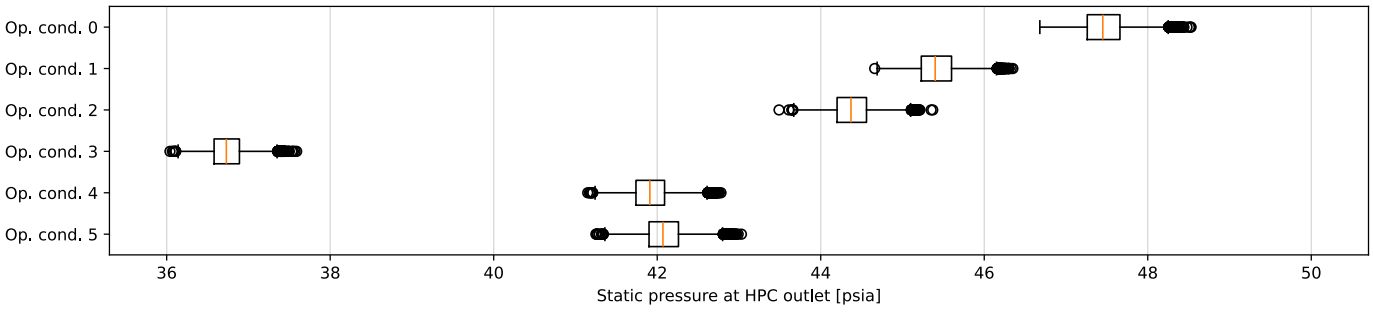

Figure 6: Box plot of static pressure at HPC outlet (`Ps30`) by operating condition.

The need for standardization per operational condition is evident from the graphs above. Standardization normalizes the data, making it more suitable for subsequent analysis and modeling. It also removes the external variability introduced by different operating conditions, allowing the fault characteristics to be represented more accurately. For standardization, the z-score method is applied, as shown in Equation 2, where $\mu_{j,c}$ and $\sigma_{j,c}$ are the mean and standard deviation of sensor j under operational condition c, and c is the operational condition of observation i. These parameters are estimated from the training data of all subsets combined and then applied to both training and test data. Predictors whose standard deviation falls below 10^{-3} within any combination of subset and operational condition are dropped, as they provide no useful information.

$$x'_{i,j} = \frac{x_{i,j} - \mu_{j,c}}{\sigma_{j,c}} \tag{2}$$

Post-standardization, extreme values ($|z| > 5$) were found in less than 1.4 % of observations across all subsets. Given their small proportion and consistency with late-life degradation, no observations were removed.

Effect of standardization can be seen in Figure 7, where distinct clusters corresponding to different operational conditions are no longer present. The values are now centered around zero, as expected from the z-score standardization.

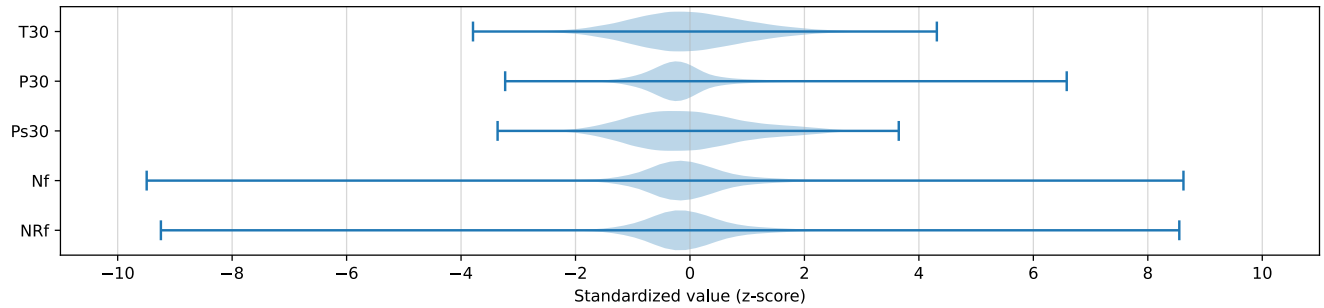

Figure 7: Violin plot of key sensors (`T30`, `P30`, `Ps30`, `Nf` and `NRf`) after standardization.

Figure 8 highlights the effect of standardization on bypass ratio (`BPR`) for two engine units in `FD004`. The two units show visibly different degradation trajectories, which may reflect the presence of two distinct fault modes in `FD004`; however, the dataset provides no per-unit fault mode label to confirm this.

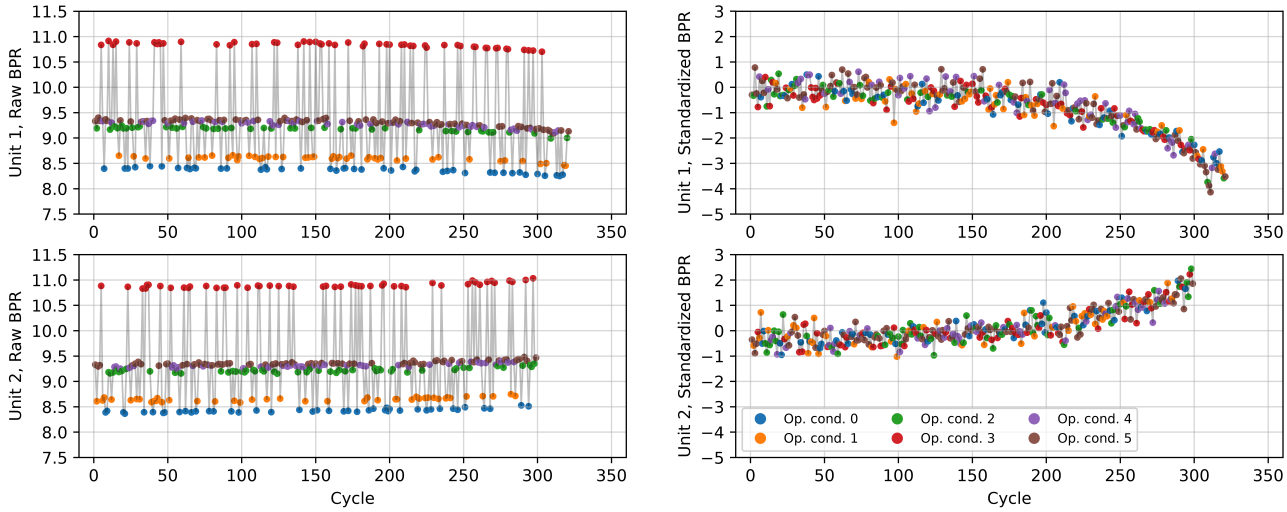

Figure 8: Bypass ratio (`BPR`) for `FD004` before standardization (on the left) and after standardization (on the right).

After standardization, a trailing rolling mean with a window of 10 cycles is applied to each sensor within each unit to reduce measurement noise. Only the current and previous cycles of the same unit are used (for the first cycles the window is shortened). The smoothing is applied identically to the test data.

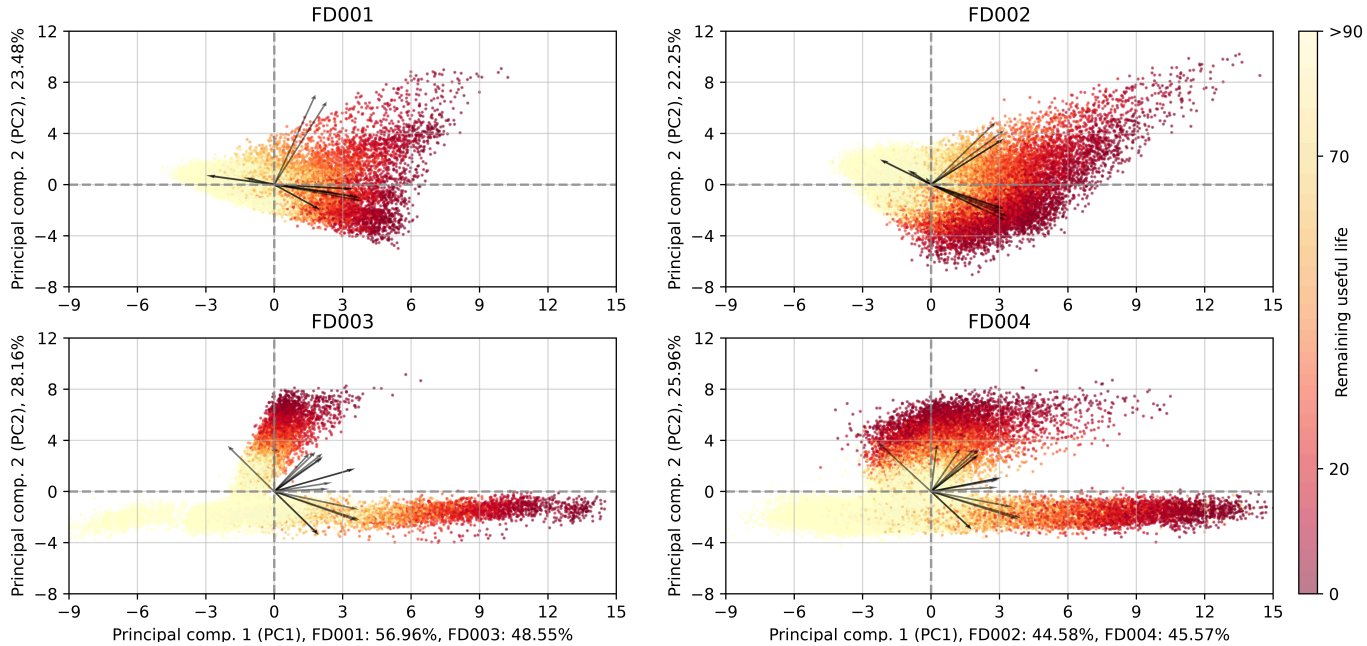

Figure 9: PCA biplots after data pretreatment for all *NASA Turbofan Jet Engine* subsets.

Finally, the effect of standardization on the PCA biplots can be observed in Figure 9. In comparison with Figure 3, there are no longer separable clusters (in case of `FD002` and `FD004` which contain multiple operational conditions). The score points in `FD001` and `FD003` show similar shape and spread, as do those in `FD002` and `FD004`, indicating that standardization has effectively removed the influence of different operating conditions. The subsets with multiple fault modes (`FD003` and `FD004`) also show a visible split in the score trajectories, which may reflect the two underlying fault modes, although this cannot be confirmed without ground-truth labels.

Modelling Plan

Modelling Goal

The goal is to calibrate a soft-sensor to estimate the remaining useful life (RUL) maintenance due time for turbofan engines using the *NASA Turbofan Jet Engine* dataset. It is hard to set rigid KPIs for the Q^2 metric or operational tolerance limits before modelling, as we do not know the exact operational tolerance limits of the engine's RUL. Moreover, if our model goes off at high-RUL values (has a large residual for high RUL – such as predicting an RUL of 120 when in reality it is 95) and predicts low-RUL very well with a very small residual, it can be considered as a good model. When an engine is close to maintenance (has a small RUL) the information is much more useful and critical.

Model Calibration Strategy

To calibrate the model, tools and methodology are applied using the *NASA Turbofan Jet Engine* training dataset. Since we have one response variable ($Y = \text{RUL}$) and not a multivariate Y, a univariate version of partial least squares (PLS) is used. To evaluate the calibration, $R^2_{\text{cal}}(X)$ and $R^2_{\text{cal}}(Y)$ can be used. The matrices presented in the calibration strategy are placed next to their respective equations:

Bi-linear decomposition: Decomposes the input matrix (X) and output matrix (Y) into score matrices (T, U), loading matrices (P, Q) and residuals (E, F) respectively, as can be seen in Equation 3.

$$\begin{aligned} X &= T \cdot P^T + E \\ Y &= U \cdot Q^T + F \end{aligned} \tag{3}$$

Score relation and covariance maximization: Establishes the relationship between the score matrix of input (T) and output (U) to maximize covariance, as can be seen in Equation 4.

$$U = T \cdot d + G \tag{4}$$

Here d is the scaling/regression coefficient linking the score components and G is the residual matrix.

Regression coefficients: Using the weight matrix (W) and loading matrices (P and Q) final regression coefficients are calculated, which will be used with the test dataset to predict the RUL of the engine, as can be seen in Equation 5.

$$b_{\text{pls}} = W \cdot \left(P^T \cdot W\right)^{-1} \cdot Q^T \tag{5}$$

Model Validation Strategy

The initial scope of the validation ensures that model is not overfitting or underfitting meaning that even outside of the training data model learns well and predict the output with accuracy. Also, deciding the number of latent variables (LV) is part of the validation. Too many LVs may lead to local model and too few LVs may not capture relevant variations. Another validation point would be to verify if the model works well for small or large values.

To tune the internal parameters, cross validation is used. The validation strategy follows below order of steps:

- When performing k-folds, it will be made sure that the units (entire engine units as small time-series) are divided and not individual samples. The units stay together in a bucket, ensuring that consecutive cycles of an engine do not get split across different folds.
- Leave out 10 %–25 % of the total units to form groups on the remaining samples, ensuring each sample group is left out only once.
- The left-out group samples are predicted by the model.
- On the basis of the summation of prediction error or maximum Q^2 value, the number of LVs with minimum error is chosen.

Model Testing Strategy

To test the model, RMSE and R^2 value can be used to determine the error and predictive performance of the model. We can also analyze the residual and how they vary as a function of the RUL. R^2 is a metric to evaluate the goodness of fit for the test data. The *NASA Turbofan Jet Engine* test dataset is completely independent of the training data, ensuring the model remains unbiased toward the train set.

Mathematical Methods Used in Modelling

Bi-linear decomposition decomposes the input matrix (X) and output matrix (Y) into score matrices (T, U), loading matrices (P, Q) and residuals (E, F) as $X = T \cdot P^T + E$ and $Y = U \cdot Q^T + F$. Score relation and covariance maximization establish the relationship between the score matrix of input (T) and output (U), maximizing the covariance through $U = T \cdot d + G$, where d represents the scaling coefficient and G is the residual. Finally, regression coefficients are calculated using the weight matrix (W) and loading matrices (P and Q) via $b_{\text{pls}} = W \cdot \left(P^T \cdot W\right)^{-1} \cdot Q^T$ to predict the RUL on the new data. A univariate version of PLS is utilized because there is only one response variable ($Y = \text{RUL}$).

Model Diagram

Model diagram is shown in Figure 10 below.

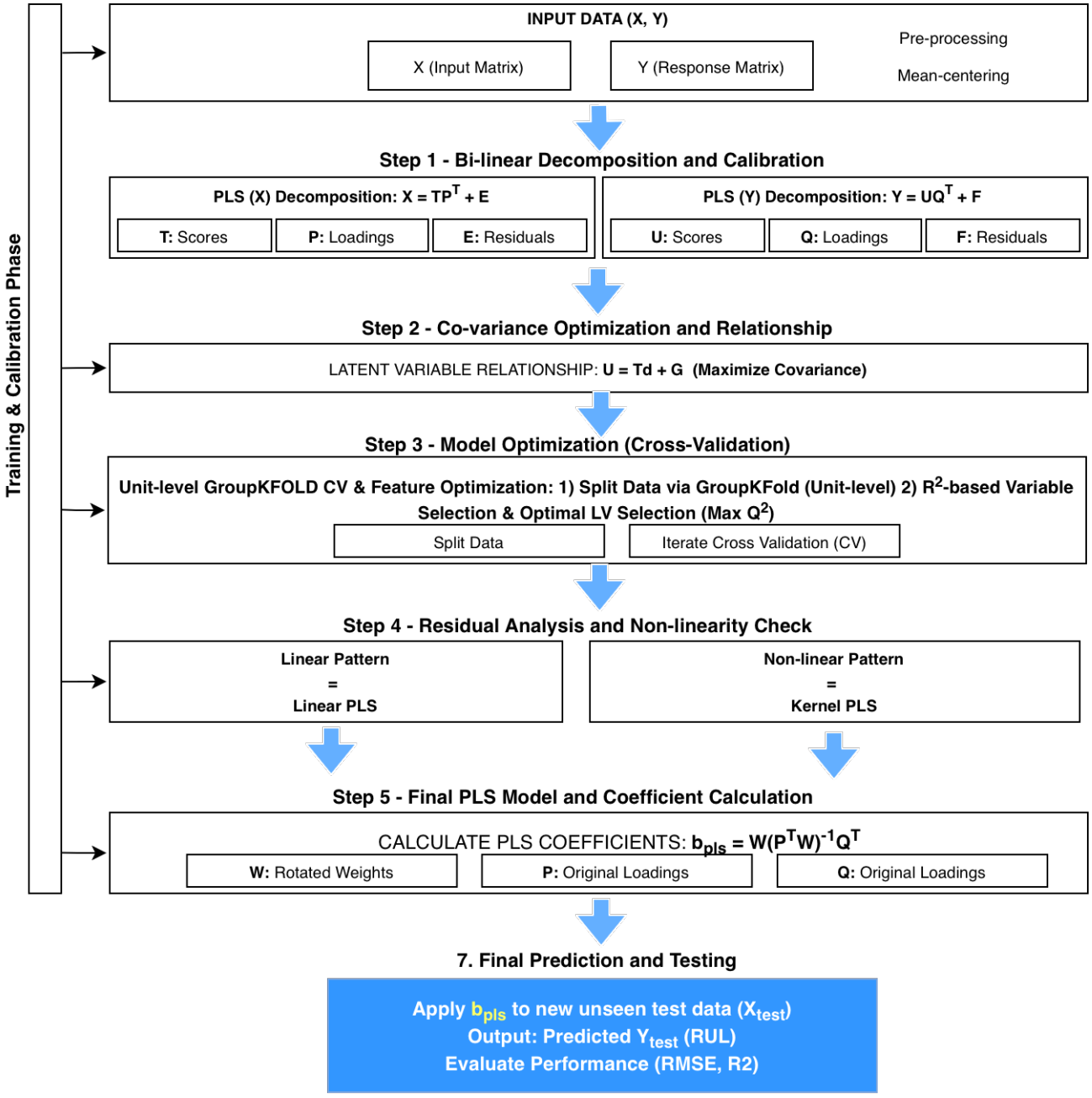

Figure 10: Flowchart of the model development process.

Results

The implementation uses `PLSRegression` which is linear PLS. In our *NASA Turbofan Jet Engine* dataset file we have 4 different subsets, all of them containing train dataset, test dataset and corresponding RUL for the test dataset, with different operational settings and failure condition. After the data pretreatment, we have used VIP and cross-validated R^2 of individual sensors to select important variables. A sensor is kept if its VIP averaged across subsets is above 0.5 (Figure 11) and its average R^2 is above 0 (Figure 12). All 15 sensors satisfied both criteria, so no further sensors were removed. After that, we have built a correlation heatmap with the selected variables (Figure 13), where it can be seen that sensors in `FD001` and `FD002` are strongly correlated with each other, while in `FD003` and `FD004` the correlations are weaker (e.g. `BPR`, `W31` and `W32`), which may reflect the two different fault modes affecting different sensors.

The final results of our four trained partial least squares models (RMSE and R^2) are shown in Table 4.

Subset	Operating condition	Fault modes	RMSE	R^2
`FD001`	Single (sea level)	Single (HPC degradation)	21.72	0.71
`FD002`	Six (discrete)	Single (HPC degradation)	20.46	0.77
`FD003`	Single (sea level)	Two (HPC deg. and fan deg.)	20.88	0.72
`FD004`	Six (discrete)	Two (HPC deg. and fan deg.)	21.95	0.74

Table 4: Partial least squares models results (RMSE, R^2) for each of the subsets.

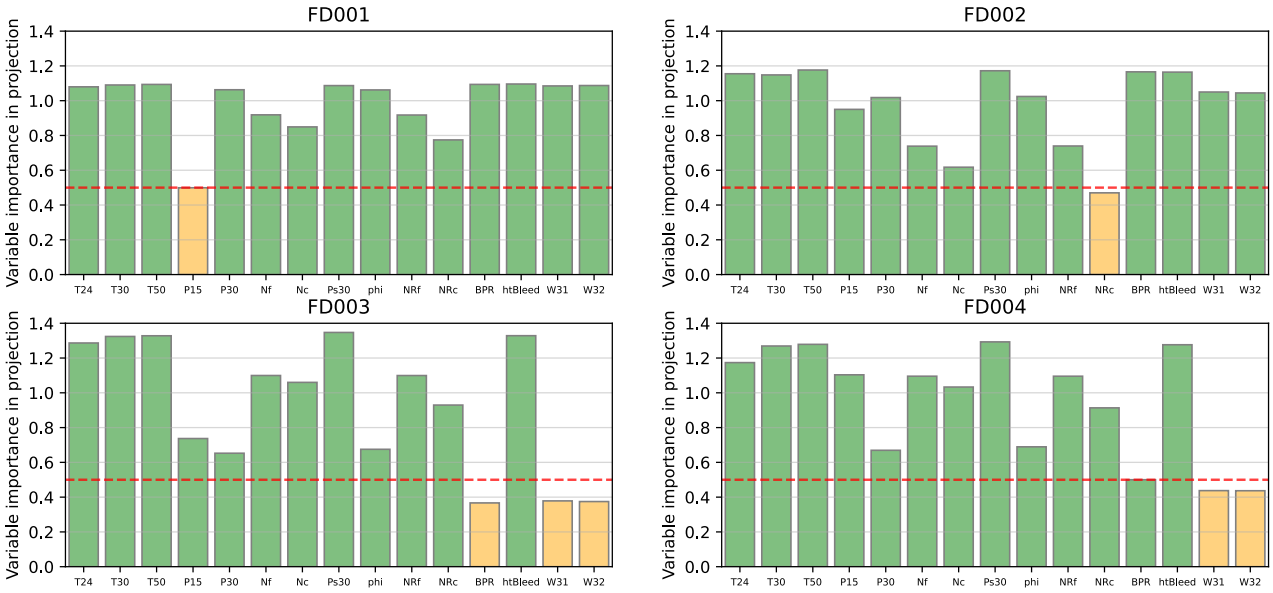

Figure 11: Variable importance in projection (VIP) across all subsets (`FD001` – `FD004`).

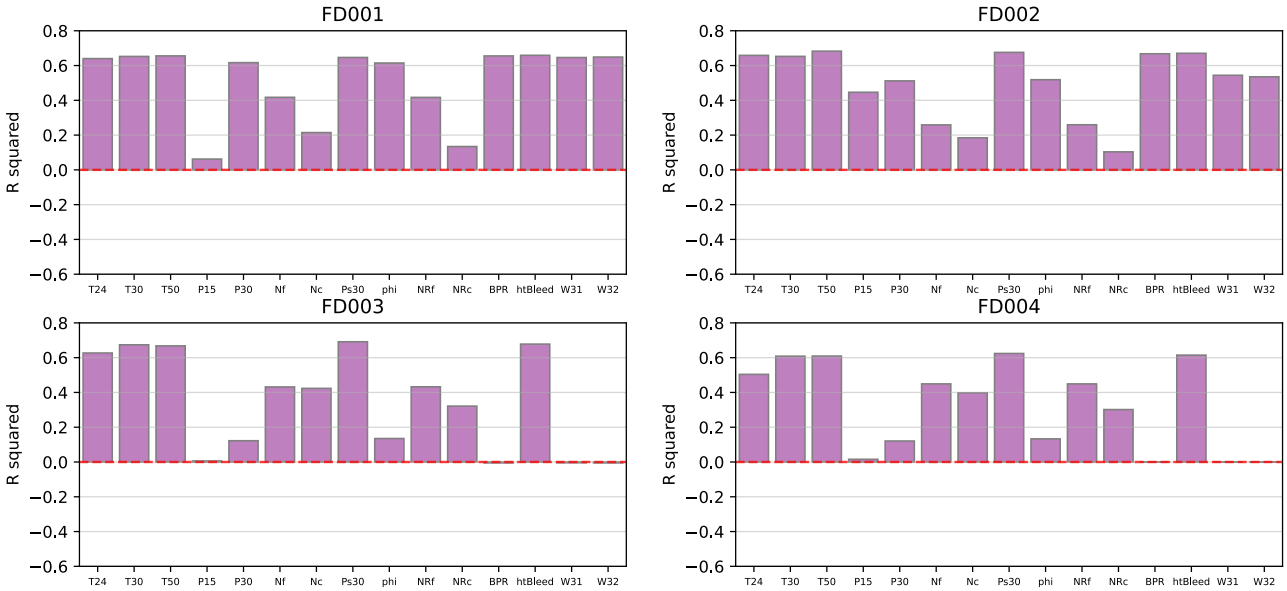

Figure 12: R^2 cross-validation across all subsets (`FD001` – `FD004`).

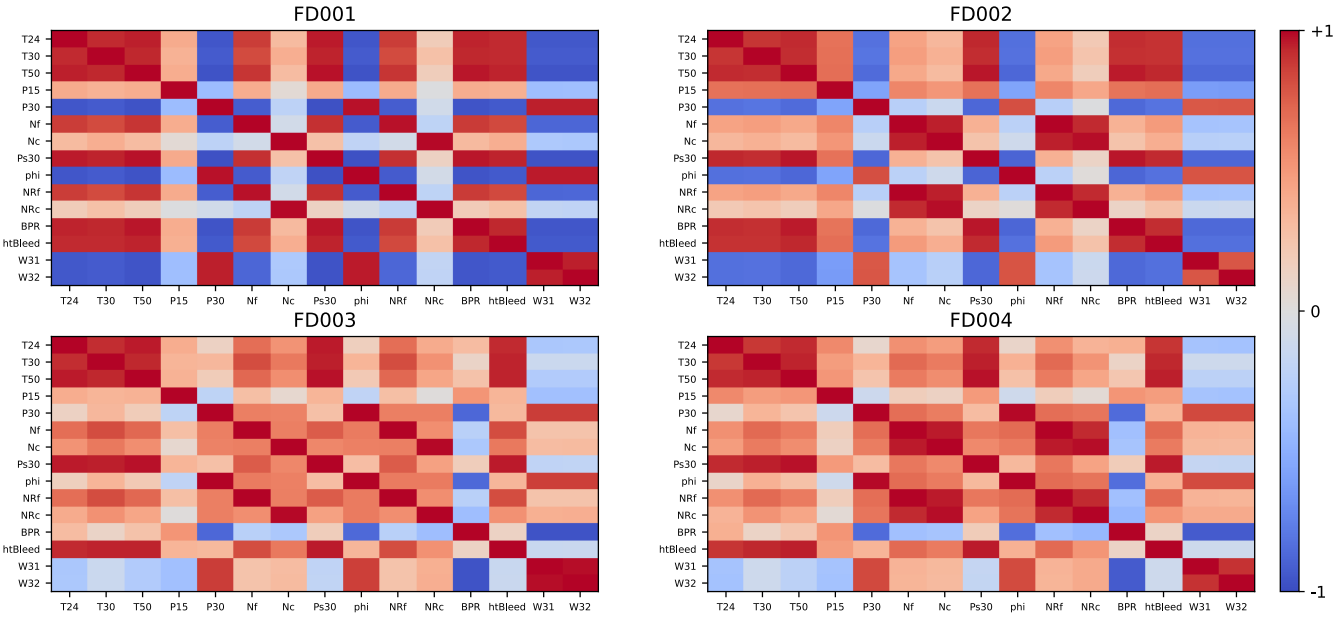

Figure 13: Sensor measurements correlation matrices for each subset.

The number of latent variables was selected based on Q^2 cross-validation (Figure 14) as the smallest number of LVs whose Q^2 is within 0.01 of the maximum. This resulted in 2, 1, 3 and 3 LVs for `FD001` – `FD004`, respectively. The selected LVs correspond to Q^2 values between 0.74 and 0.78 for all subsets, which indicates good predictive ability. Adding more LVs does not improve Q^2 noticeably. The R^2 on the test data (Figure 15) is shown only for comparison and to check whether the models are overfitting or not. Similarly to Q^2, the R^2 values span from 0.71 to 0.77 for the selected number of LVs, which is close to Q^2, indicating that the models are not overfitting.

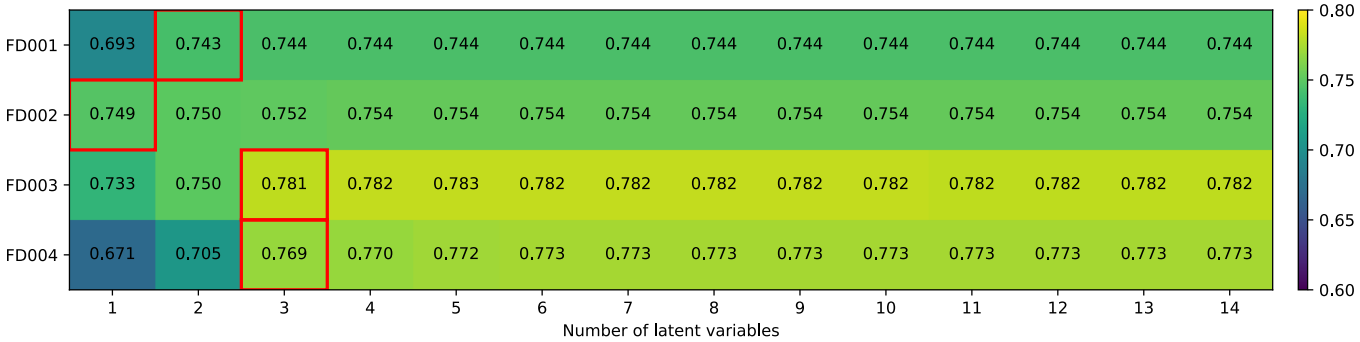

Figure 14: Q^2 performance across all subsets based on number of LVs (red box shows chosen LVs).

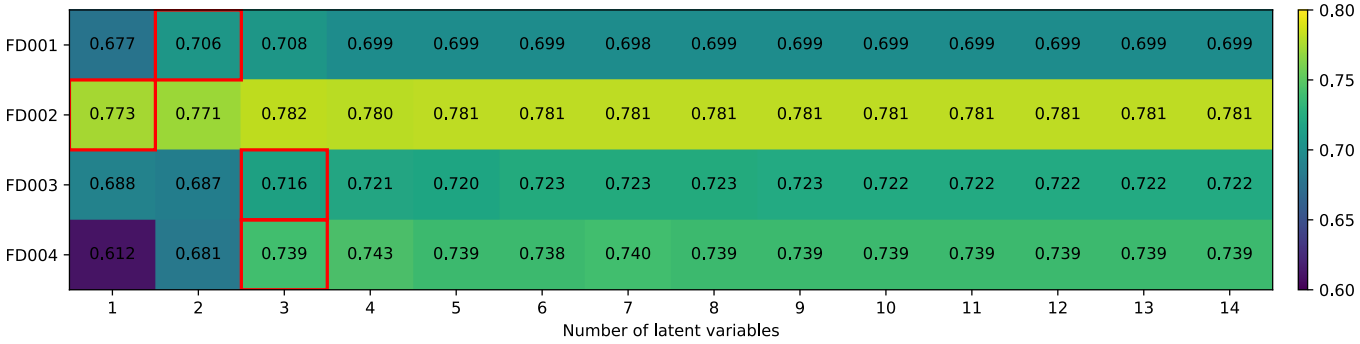

Figure 15: R^2 performance across all subsets based on number of LVs (red box shows chosen LVs).

Figure 16 shows the comparison of predicted and true remaining useful life. Our trained models follow the diagonal (highlighted in red) reasonably well. As previously stated, an RUL of 125 corresponds to RUL ≥ 125, meaning a healthy state of the engine unit. For these units the predictions in `FD002` and `FD004` spread between approx. 75 and 125. Overall, our models tend to overestimate the RUL mainly for RUL lower than 80, which is worse than underestimating it, as the maintenance could be planned too late.

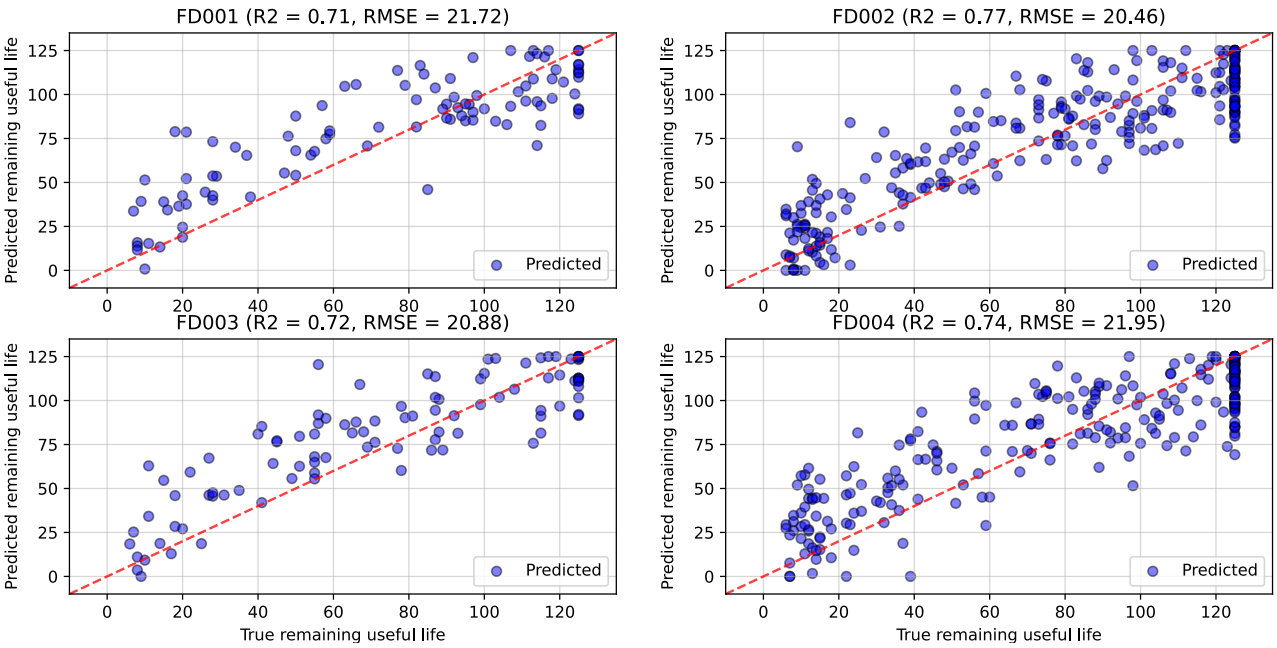

Figure 16: Predicted vs observed remaining useful life across all subsets (`FD001` – `FD004`).

Residual analysis shown in Figure 17 provides similar information as Figure 16, showing that the residuals are within approximately ± 60 cycles for all subsets. The diagonal lines of points in `FD002` and `FD004` are caused by the introduced RUL capping. The residuals also show a slight systematic pattern (overestimating predictions for RUL $\in [20, 80]$), suggesting that the relation between sensor measurements and RUL is not fully linear. A non-linear PLS (e.g. kernel PLS) could therefore improve the predictions.

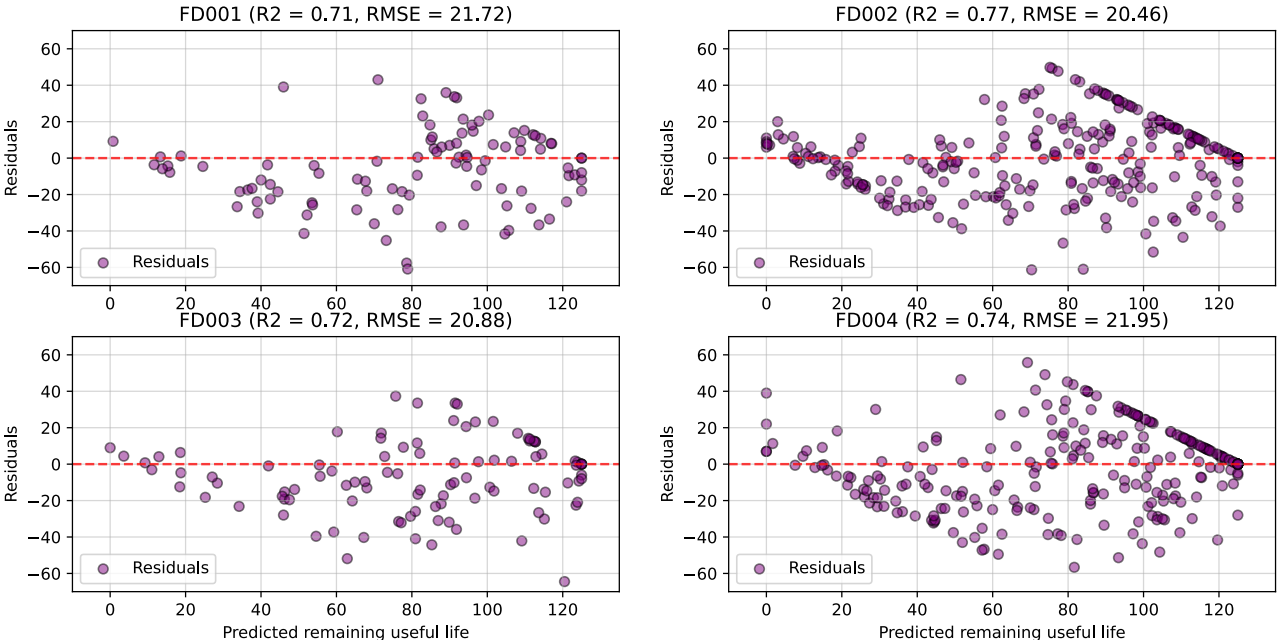

Figure 17: Residual analysis across all subsets (`FD001` – `FD004`).

Appendix

Symbol	Description	Unit
unit_id	Unit number	—
cycle	Time	cycle
altitude	Altitude	K ft
mach	Mach number	—
throttle	Throttle resolver angle	—
T2	Total temperature at fan inlet	°R
T24	Total temperature at LPC outlet	°R
T30	Total temperature at HPC outlet	°R
T50	Total temperature at LPT outlet	°R
P2	Pressure at fan inlet	psia
P15	Total pressure in bypass-duct	psia
P30	Total pressure at HPC outlet	psia
Nf	Physical fan speed	rpm
Nc	Physical core speed	rpm
epr	Engine pressure ratio (P50/P2)	—
Ps30	Static pressure at HPC outlet	psia
phi	Ratio of fuel flow to Ps30	pps/psi
NRf	Corrected fan speed	rpm
NRc	Corrected core speed	rpm
BPR	Bypass ratio	—
farB	Burner fuel-air ratio	—
htBleed	Bleed enthalpy	—
Nf_dmd	Demanded fan speed	rpm
PCNfR_dmd	Demanded corrected fan speed	rpm
W31	HPT coolant bleed	lbm/s
W32	LPT coolant bleed	lbm/s

Table 5: *NASA Turbofan Jet Engine* dataset columns (all of them) and their description.